Microsoft



# Microsoft Build 2017

Microsoft

P4048

# What's New in Xamarin.Forms

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#MSBuild





photo credit: stltoday.com

# Xamarin Native approach

iOS

iOS C#

Android C#

Windows C#

Shared C# Logic

3 Native User Interfaces

Shared App Logic



# Xamarin.Forms approach





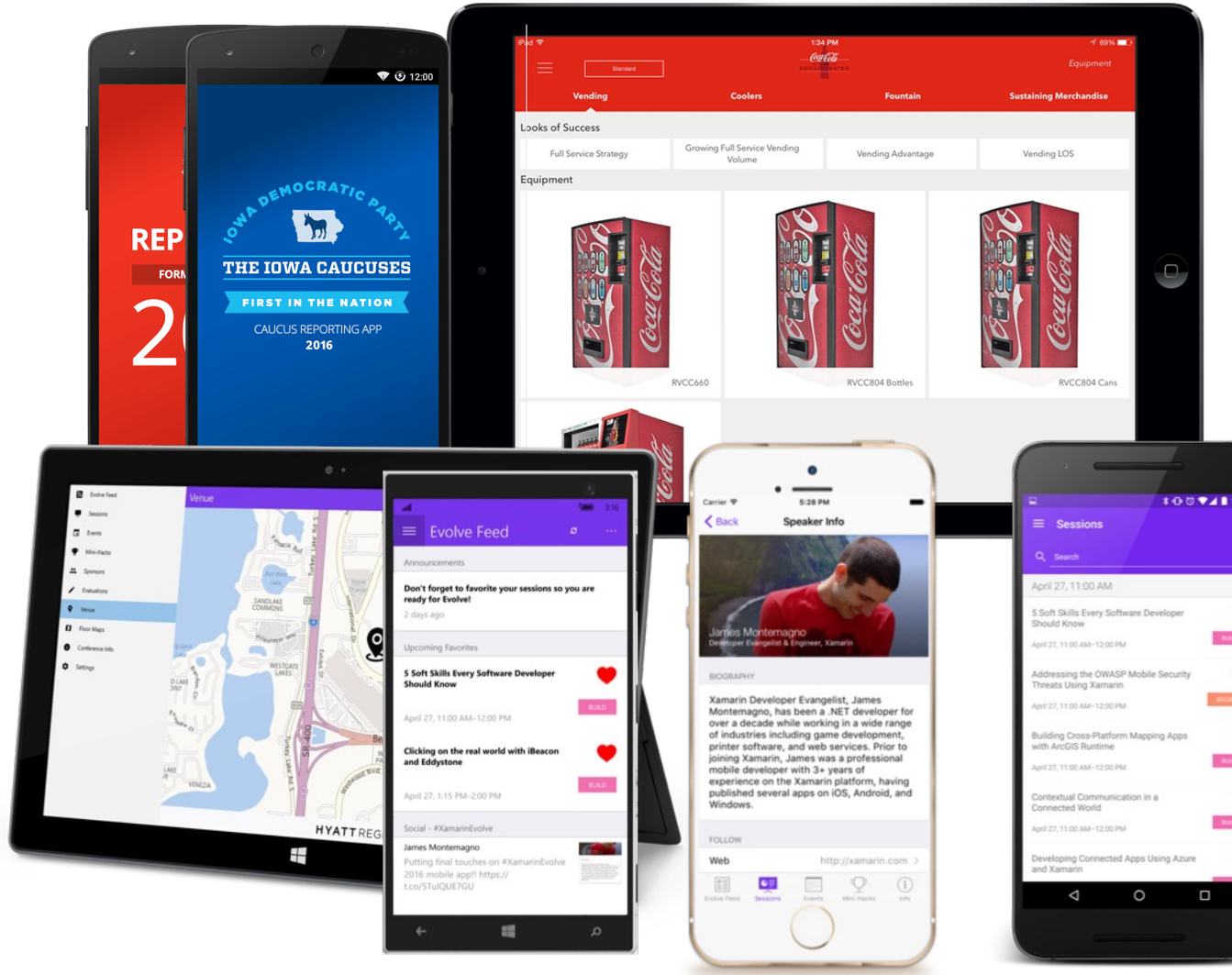



Shared UI Code

Shared C# Logic

Shared (Native) User Interfaces

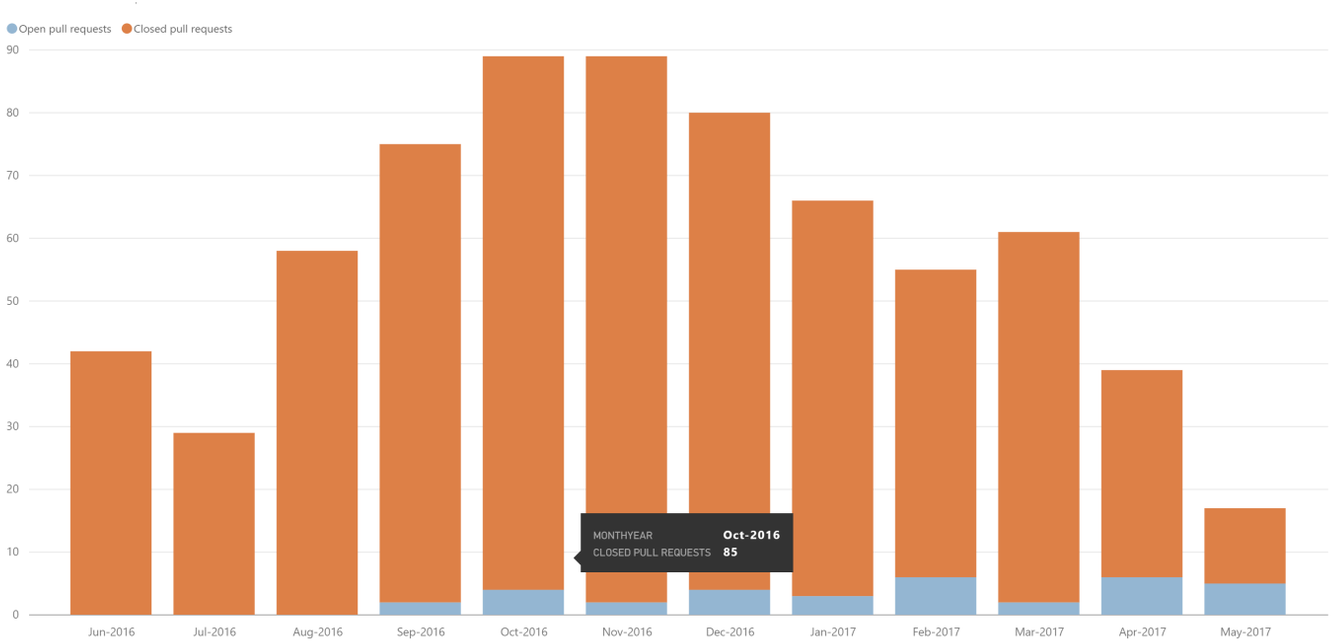
Shared App Logic



# 1 Year of Open Source!

# Community

- Public Roadmap: aka.ms/xfroadmap
- 70 contributors
- 700 merged pull requests
- 37 evolution proposals



# Features and Enhancements

Fast Renderers Preview

Memory Improvements

Native Control Binding

Bindable Picker

Map Extensibility

XAMLC

Startup Time

Forms Previewer

Android Support Libraries

FlexLayout Preview

Frame Corners

Enhanced GestureRecognizers

Tizen

Blur Behind

macOS Preview

Themes

ListView Scrolling

Embedded Native Controls

Deep Linking

.netstandard support

Fast Scroll

Styles

NavigationPage Translucency

Xamarin.Forms Embedding

PlatformSpecifics

Accessibility

Partial Collapse MasterPage

DataPages

# Controls and Tools Ecosystem

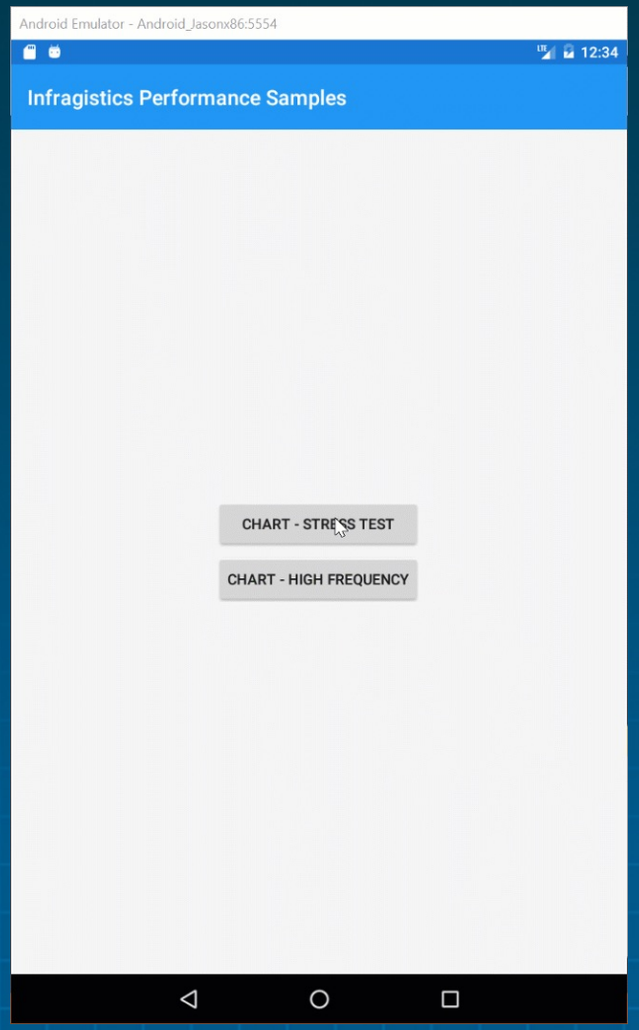
- Forms Community Toolkit
- GrapeCity
- Infragistics
- mfractor
- Syncfusion
- Telerik
- The Xamarin Show | Channel 9
- Xamarin Podcast - xamarinpodcast.com

# Infragistics Ultimate UI for Xamarin

**The complete toolkit for building fast, beautiful, native mobile apps**

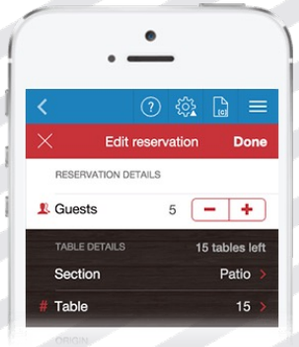
- Lightning-fast UI controls
  - Data grids gracefully render even the heaviest data loads
  - Data charts that handle millions of data points with millisecond refresh rates
- Revolutionary WYSIWYG design tools for Xamarin.Forms

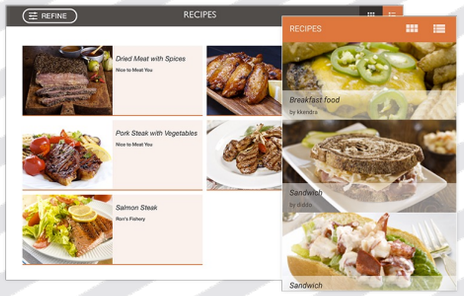


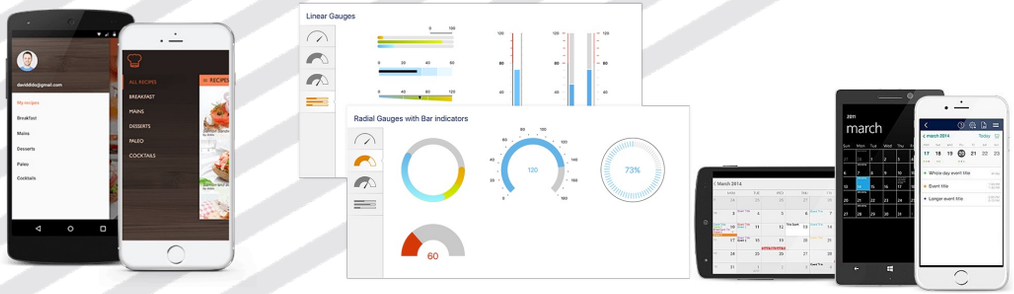

**Free trial: www.infragistics.com/xamarin**
**20% off for Build attendees through July 15 – Use promo code MSBUILD17**

# Why Telerik UI for Xamarin




1. Polished UI controls out of the box
2. Allows developers to focus on app features, rather sweating UI details
3. Native controls for Xamarin.iOS & Xamarin.Android
4. Cross-platform wrappers for Xamarin.Forms
5. Extreme performance tuned
6. Works seamlessly on Windows/Mac
7. Easy integration through NuGet and Templates
8. Lot of customization options
9. Sample apps in Stores







The Best .NET UI for .NET Ninjas!

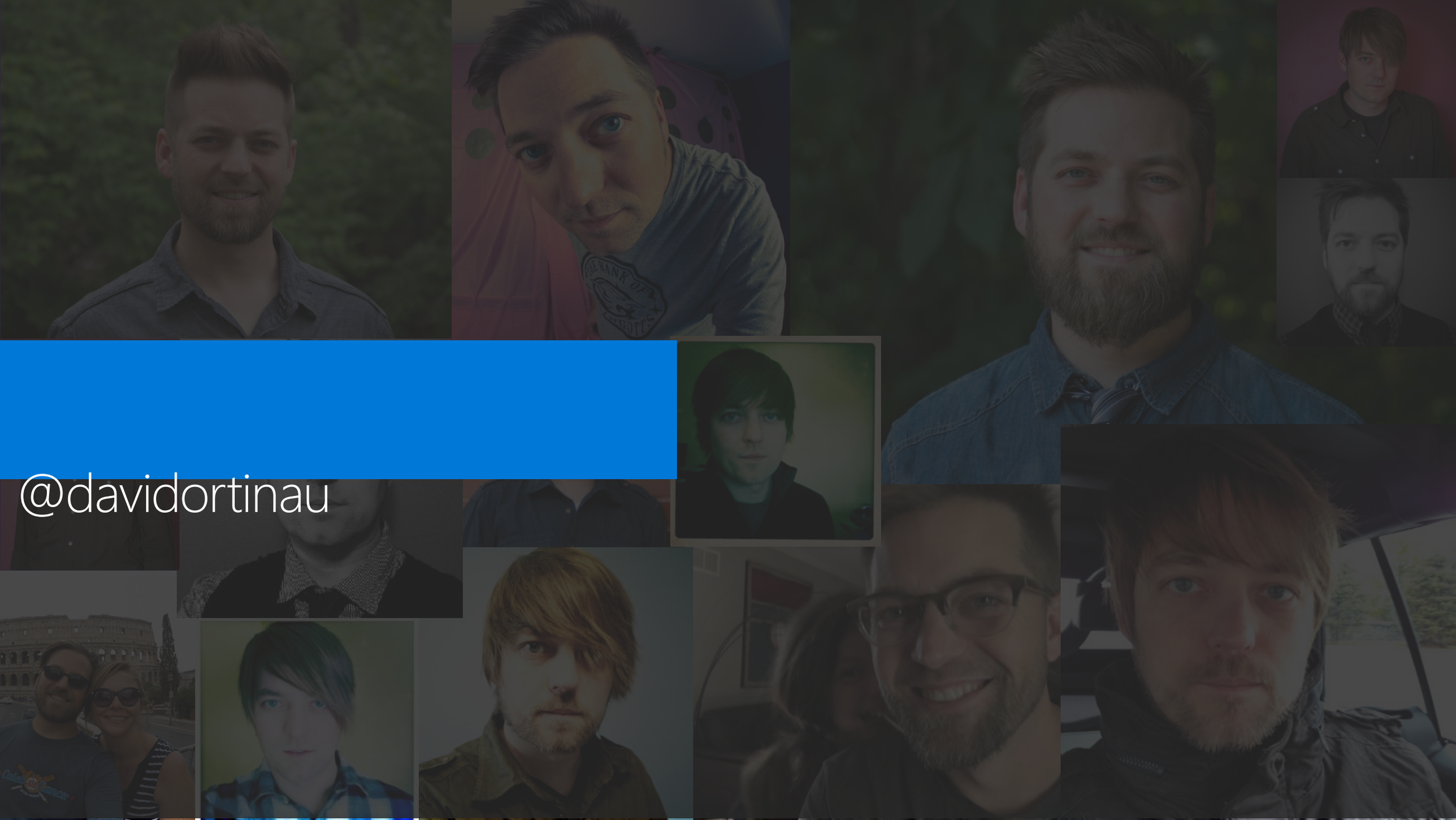


telerik.com/xamarin-ui

# Quality and Performance

- 2.2.0 last //build compared to 2.3.5
- Aggressive bug report triage
- Android Performance Improvements
  - Startup
  - Fast Renderers
  - XAMLC

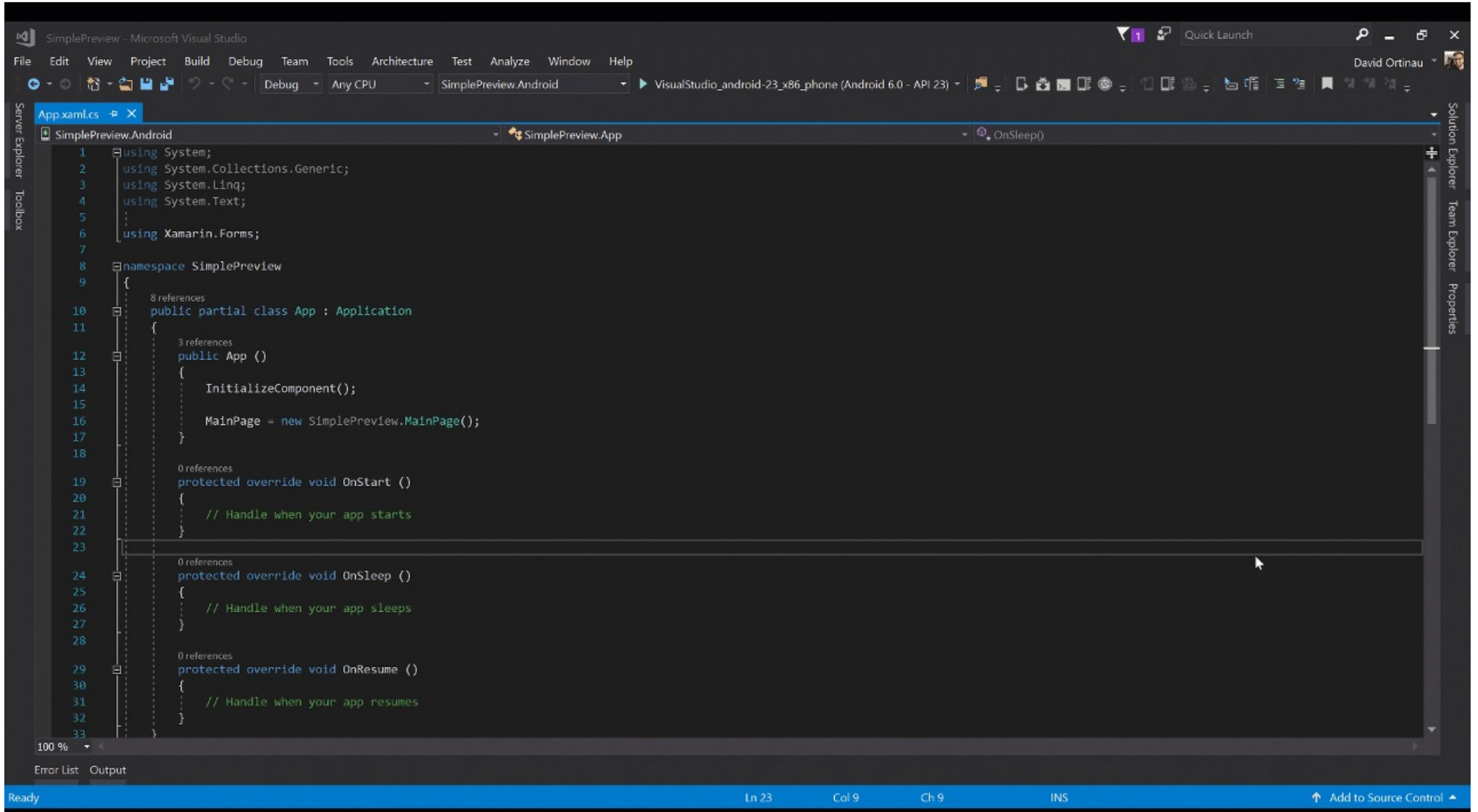
# What have you done for me lately?



@davidortinau

# Xamarin.Forms Previewer

- Visual Studio for Mac & Visual Studio 2015 / 2017



# Available Today: Xamarin.Forms 2.3.5-pre3

- .NET Standard Compatible
  - Recommend 1.1+ target
- Fast Renderers for Android
  - Button, Image, Label
- Accessibility
- macOS Preview
- XAMLC type improvements
- 160+ bug fixes and enhancements

# Demo
# Platform Specific Controls
with Live Player

# The Future:
# Xamarin.Forms 3.0

# FlexLayout

- A FlexBox inspired layout system
- What's it good for?
  - Flowing items
  - Adaptive layout
- What is supported?

# FlexLayout Example

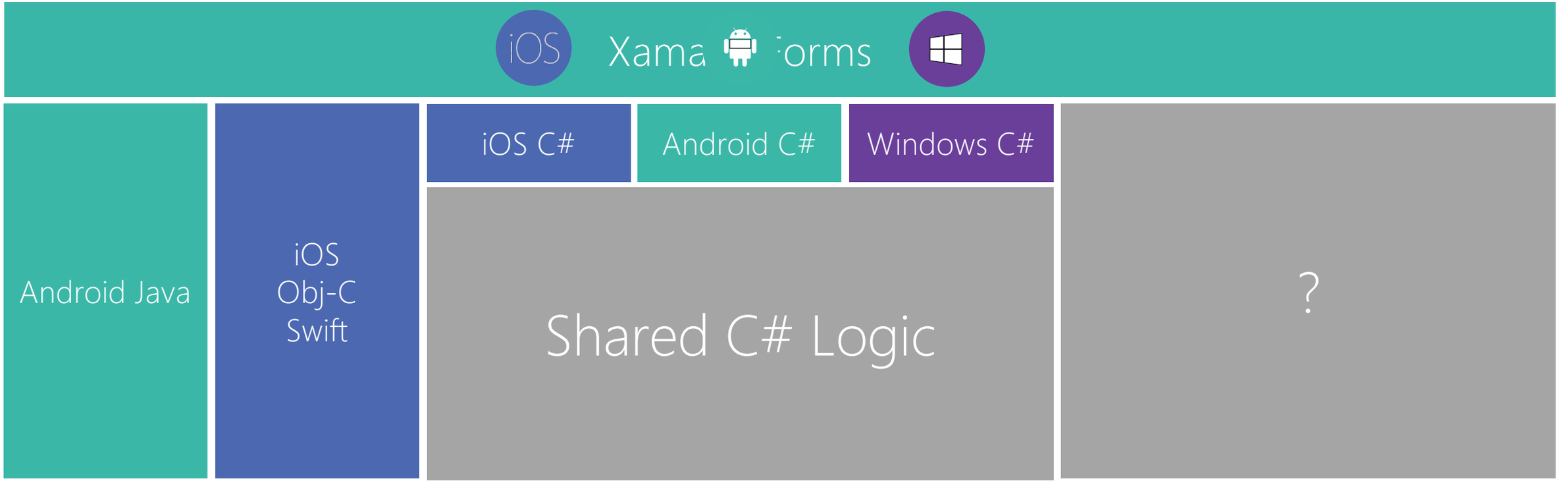
```
<?xml version="1.0" encoding="UTF-8"?>
<ContentPage x:Name="rootElement" Title="FlexibleBox" xmlns:flex="clr-namespace:Xamarin.Forms;assembly=Xamarin.Forms.Flexlayout"
xmlns="http://xamarin.com/schemas/2014/forms" xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
x:Class="FormsFlexLayoutDemo.FlexDemoPage">
    <ContentPage.Resources>
        <ResourceDictionary>
            <Style x:Key="FlexLayoutStyle" TargetType="flex:FlexLayout">
                <Setter Property="Overflow" Value="{Binding SelectedOverflow}" />
                <Setter Property="AlignItems" Value="{Binding SelectedAlignItems}" />
                <Setter Property="AlignContent" Value="{Binding SelectedAlignContent}" />
                <Setter Property="AlignSelf" Value="{Binding SelectedAlignSelf}" />
                <Setter Property="FlexDirection" Value="{Binding SelectedFlexDirection}" />
                <Setter Property="Wrap" Value="{Binding SelectedWrap}" />
                <Setter Property="JustifyContent" Value="{Binding SelectedJustifyContent}" />
                <Setter Property="Padding" Value="10" />
                <Setter Property="BackgroundColor" Value="#FFD54F" />
            </Style>
            <Style x:Key="DemoItemStyle" TargetType="ContentView">
                <Setter Property="WidthRequest" Value="{Binding Source={x:Reference rootElement}, Path=BindingContext.ItemWidth}" />
                <Setter Property="Margin" Value="10" />
                <Setter Property="Padding" Value="5" />
                <Setter Property="BackgroundColor" Value="White" />
            </Style>
        </ResourceDictionary>
    </ContentPage.Resources>
    <Grid>
        <Grid.RowDefinitions>
            <RowDefinition Height="40" />
            <RowDefinition />
```

# Layout Compression

- ?

# Embedding

- ContentPages, DependencyService, MessagingCenter



# New Platforms

- Samsung Tizen
  - Televisions, Wearables, Mobile
- macOS
- GTK#

# Demo
# FlexLayout

# Demo
# Xamarin.Forms Embedding

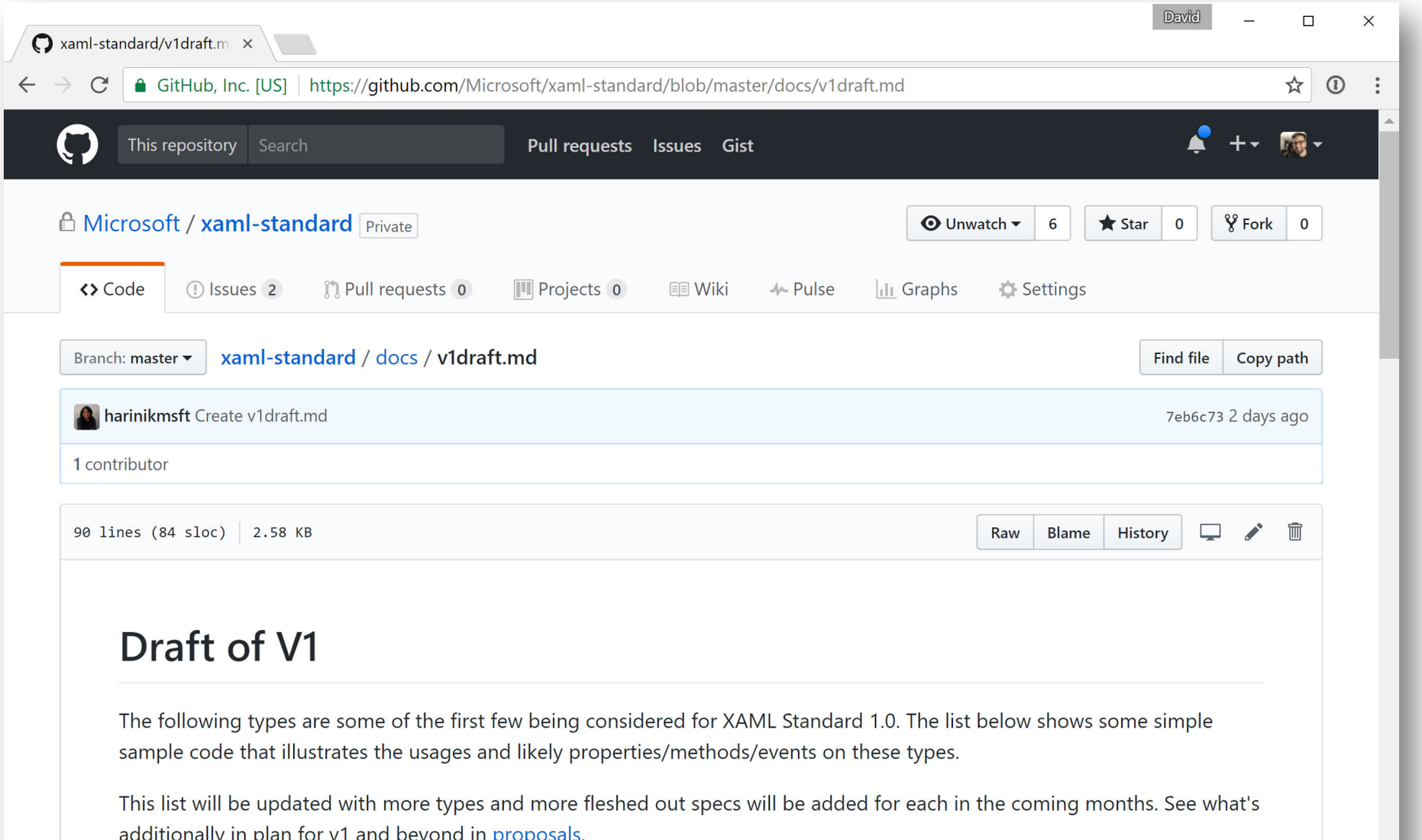
# Demo
# macOS
# GTK#

# Also to be included

- Fast Renderers
- ListView improvements
- OneTime binding - big performance improvement
- XAMLC – reducing reflection and runtime lookups
- CSS-Like Styling
- New Navigation concepts - Master Detail - new, proper hierarchy
- Visual State Manager
- Better adaptive layout - mirror UWP relative layout
- Standardization of renderer APIs

# XAML Standard

- github.com/Microsoft/xaml-standard



# What's Next?

Download Xamarin.Forms 2.3.5-pre3 today!
Watch for 3.0 Pre-release soon.
www.xamarin.com/download

# Thank you.
# Questions?


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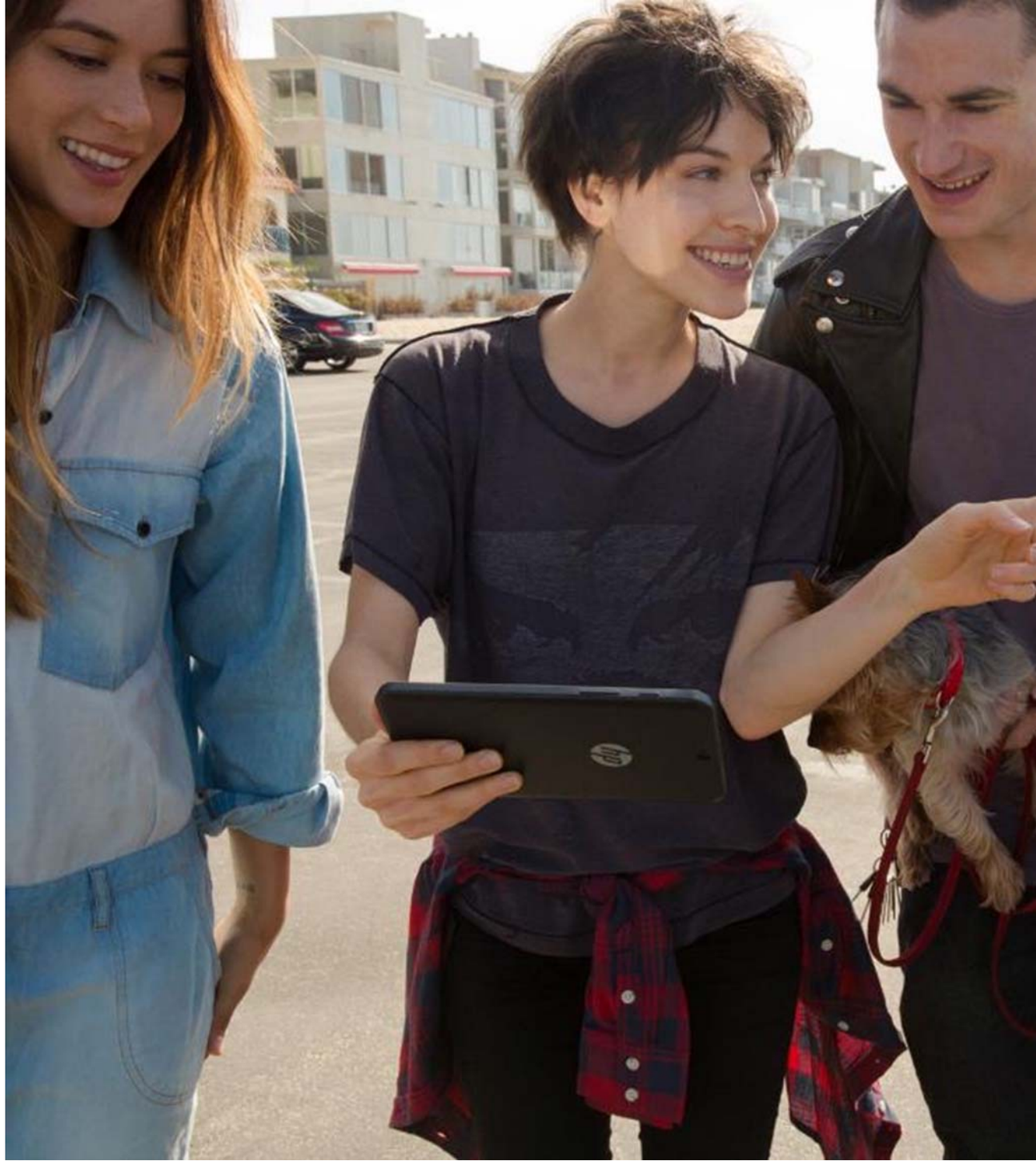
Jason
Smith
Engineering Team Lead, Microsoft

@jassmith87

Microsoft

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# Photo layout 1



# Software code slide

```
This slide layout uses Consolas, a monotype font
which is ideal for showing software code.
```